

Harsh Desai

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Mechanical engineer with hands-on experience in scientific instrumentation, electromechanical test stands, vacuum/HV systems, plasma diagnostics, DAQ automation, and mechanical design; experienced in taking experimental hardware from setup and integration through testing and troubleshooting.

EDUCATION

University of Michigan

Master of Engineering, Space Systems Engineering | GPA: 3.79/4.0

Coursework: Plasma & Fields Instrumentation; Plasma Measurement Techniques; Control for Aerospace Vehicles; Space Systems Design & Mgmt.

Ann Arbor, MI, USA

Aug 2024–Dec 2025

Vellore Institute of Technology

Bachelor of Technology, Mechanical Engineering

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, Fusion 360, Autodesk Inventor, AutoCAD, ANSYS Mechanical/Fluent

Instrumentation & Diagnostics: Langmuir probe, emissive probe, RPA, CEM/ESA calibration, SIMION, SRIM, Amptek DPPMCA

DAQ & Controls: Python, LabVIEW (CLAD prep), PyVISA/PyMeasure, Keithley DAQ/SMU, TDK Lambda PSUs, GPIB, serial, HV switching

Digital Readout: ADC/FPGA workflow, Zmod ADC 1410, Zynq-7000, AMD Vivado, MATLAB HDL Coder

Lab & Documentation: Vacuum chamber operation (CTI Hi-Vac / roughing to ~1 mTorr), 1.3 kV test stands; Microsoft Word, Excel, PowerPoint; technical test plans and reports

ENGINEERING & INSTRUMENTATION EXPERIENCE

Climate and Space Sciences and Engineering, University of Michigan

Ann Arbor, MI, USA

Research Assistant, LVACCS Instrumentation & Test-Stand Characterization

Feb 2026–Present

- Built and operated an instrumented 1.3 kV HV test workflow with discharge-box FMEA, Python/PyVISA ignition control, PSU logging, and DAQ synchronization; cut manual steps 15→5 and achieved 98.6% synchronized data capture
- Developed a Python/PyVISA + Tkinter operator GUI to automate ignition sequences, mass-flow control, and remote PSU coordination for high-throughput test-stand operation
- Operated Langmuir, emissive, and retarding-potential probes during vacuum-chamber experiments, performing bias sweeps and interpreting I–V / plasma-potential-related measurements across regolith-lofting test states
- Configured Keithley DAQ and TDK Lambda HV sequencing for hollow-cathode discharge testing (1.3 A constant-current, 1300 V headroom, xenon ignition near 500 V / 50 SCCM) on a CTI Hi-Vac / roughing-pumped chamber

Solar and Heliospheric Research Group, University of Michigan

Ann Arbor, MI, USA

Student Research Assistant, SSD Instrumentation Readout Chain

Jan 2025–Dec 2025

- Developed and verified an ADC/FPGA readout architecture for a solid-state detector, sizing digitization requirements and evaluating a 14-bit, 125 MSPS Zmod ADC / Zynq-7000 path using MATLAB and Vivado

Space Physics Research Lab, University of Michigan

Ann Arbor, MI, USA

Summer Intern, TestBedz Spacecraft Qualification Platform

May 2025–Jul 2025

- Built a full-stack requirement-flowdown platform for TVAC, vibration, and EMI test submissions, matching spacecraft hardware profiles to facility instrumentation limits across 6 test-type workflows
- Encoded hardware metadata, facility operating envelopes, and acceptance criteria into machine-readable compatibility checks and traceable test plans for repeatable qualification workflows
- Surfaced incompatible TVAC/vibration requests before scheduling by translating facility limit envelopes into automated submission checks

L3Harris and University of Michigan

Ann Arbor, MI, USA

Communications Lead, Uranian Orbiter and Probe Mission Study

Sep 2024–May 2025

- Led Uranian mission communications architecture trades and DSN/fault-tolerance requirement flowdown, managing 15 traceable requirements and supporting a 16% (\$450M) cost reduction through an orbiter-only configuration trade

MECHANICAL DESIGN & ENGINEERING PROJECTS

CANSAT 2022 (NASA & AAS) | Ranked 7th worldwide out of 42 teams

Aug 2021–Jul 2022

- Designed and built a 10 m unidirectional tether-deployment mechanism using a DC motor and custom worm-gear spool for non-backdrivable load transfer and controlled descent of a 2 mm braided-nylon payload line
- Integrated payload-release and camera-stabilization hardware—elastic-lid release with torsion-spring latch plus a dual-servo 2-axis gimbal—and iterated mechanical fit, assembly, and field checkout for fixed 45° imaging

Space Instrumentation Calibration & Ion-Optics Series

Jul 2025–Dec 2025

Graduate Course: SPACE 571 | Supervisor: Prof. Stefano Livi

- Calibrated Channel Electron Multiplier (CEM) response and the charge-injection → CSA/shaper → MCA signal chain, extracting gain, centroid, FWHM, and bias-dependent count response
- Calibrated Cylindrical Electrostatic Analyzer (ESA) performance from lab energy-angle scans and SIMION/SRIM ion-optics models, quantifying transmission, angular acceptance, and energy resolution
- Modeled SIMION dual-Einzel beam expansion and Bessel-box filter response, and used SRIM TRANSMIT to estimate carbon-foil energy loss / straggling against TOF-derived exit energies

Spaceport America Cup / IREC | Placed 23rd globally and 5th in Asia-Pacific

Apr 2020–Jul 2021

- Launched a Mach 0.9 solid-motor sounding rocket to a 10,000 ft target altitude with the competition team
- Developed an SRAD trajectory simulator using CFD-backed aerodynamic data for flight prediction and design iteration

CANSAT 2021 (NASA & AAS) | Placed 13th globally and 7th in Asia-Pacific

Aug 2020–Jul 2021

- Built MATLAB telemetry / ground-control software for dual maple-seed payload deployment monitoring, and modeled mono-wing descent with Blade Element Theory plus transient CFD